

Agenda

01

Dataset Overview & Preprocessing

Data cleaning pipeline, feature engineering

02

Descriptive Analytics

Readmission distribution, demographics, numerical summaries

03

Association Rule Mining

Apriori algorithm — key rules and patterns

04

Clustering Analysis

DBSCAN, Hierarchical, K-Means, GMM

05

Classification Analysis

k-NN, Random Forest, Naive Bayes, Decision Tree, SVM

06

Conclusions

Key findings, limitations, future work

01 Dataset Overview & Preprocessing

101,766

Original Rows

101,385

After Cleaning

49

Columns

3

Target Classes (NO, >30, <30)

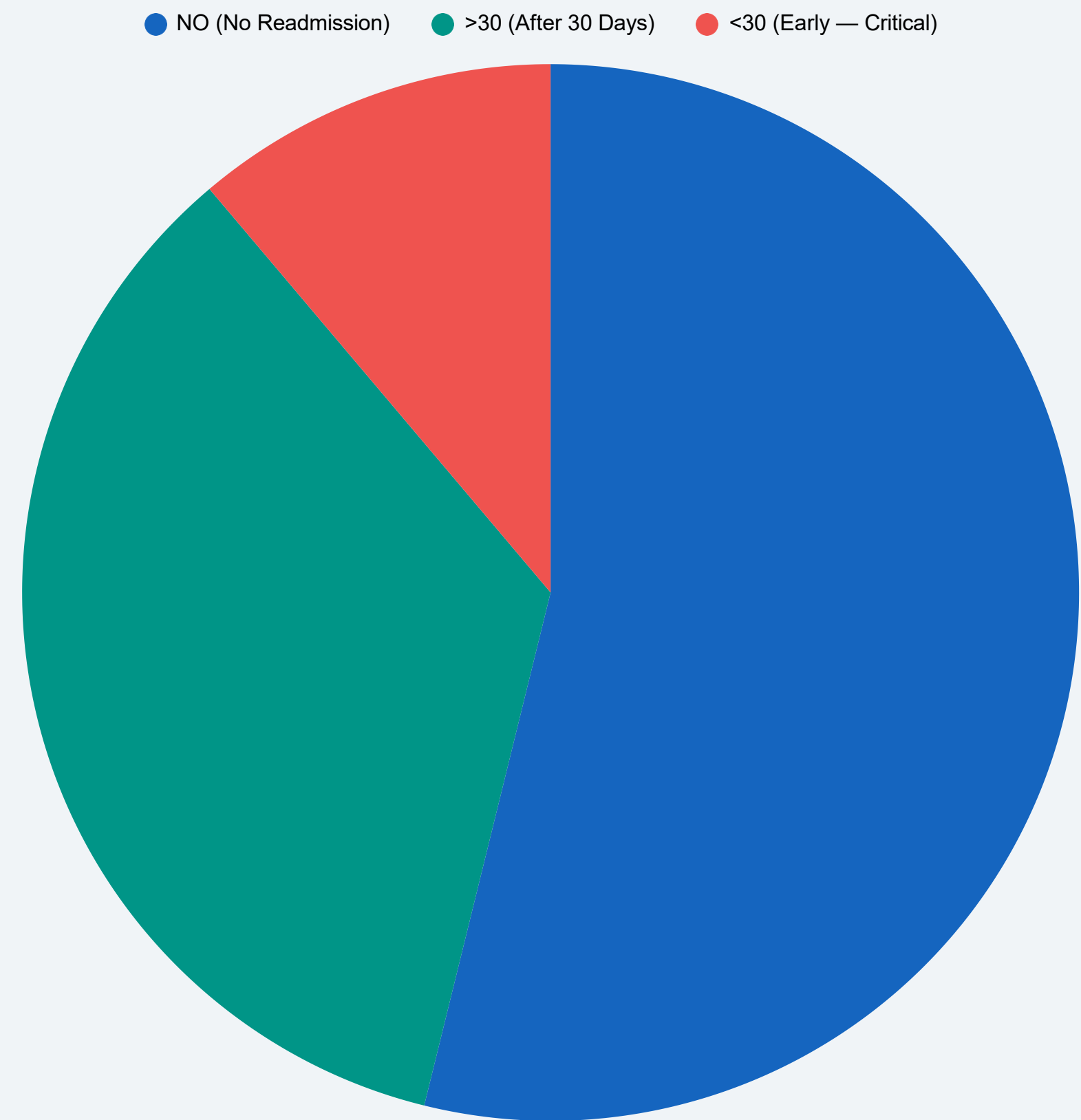
Preprocessing Pipeline

- weight column removed (96.86% missing)
- 3 rows with Unknown/Invalid gender removed
- payer_code, medical_specialty, diag_3 → 'missing'
- 1,764+ race missing values filled by class distribution
- Rows with missing diag_1 / diag_2 removed
- Numerical features discretized for Apriori analysis
- Identifier columns (encounter_id, patient_nbr) excluded

Missing Value Summary

Feature	Missing %	Action
weight	96.86%	Removed
medical_specialty	49.08%	'missing'
payer_code	39.56%	'missing'
race	2,273 rows	Distribution fill
diag_3	1.4%	'missing'

02 Descriptive Analytics — Readmission Distribution



⚠ **Clinical Priority: <30 Class**

Only 11.17% of encounters — yet represents potentially preventable short-term readmissions within 30 days. Class imbalance must be addressed in all predictive modeling phases.

Numerical Feature Summary

Feature	Mean	Max
time_in_hospital	4.40	14
num_lab_procedures	43.10	132
num_medications	16.05	81
number_inpatient	0.64	21
number_diagnoses	7.44	16

02 Descriptive Analytics — Demographic Patterns

Race Distribution

Caucasian & AfricanAmerican dominate. Dataset is demographically imbalanced — an important consideration when generalizing model results.

Gender

Female encounters slightly more frequent; distribution relatively balanced overall across the dataset.

Age

Concentrated in [60–70), [70–80), [80–90) — older diabetic patients dominate. This reflects real-world diabetes prevalence patterns.

Admission Source

Emergency room & physician referral are the two major admission pathways.

03 Association Rule Mining (Apriori)

0.2
min_support

0.7
min_confidence

1.3
min_lift

40
Filtered Rules

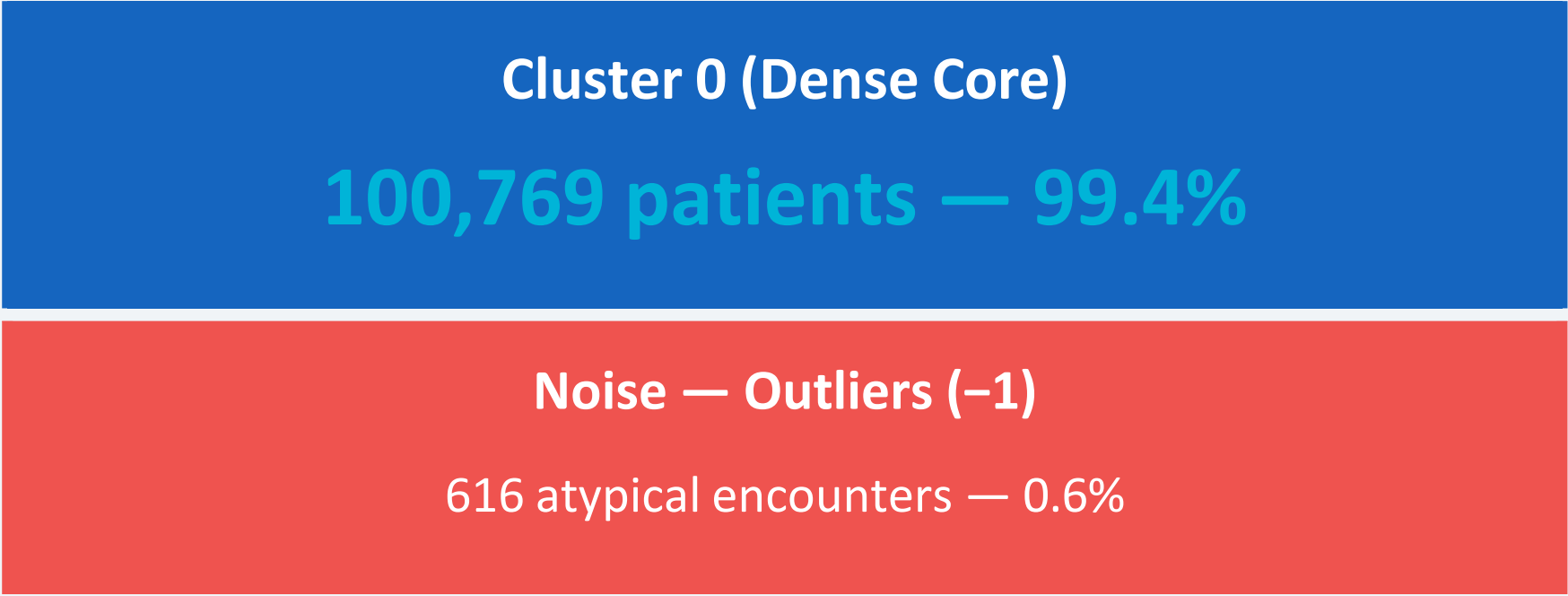
Antecedent	Consequent	Conf.	Lift
No diabetes med + no prior ER visits	Med. change not changed	100.0%	1.86
Emergency admission + many diagnoses (9+)	ER admission source	92.8%	1.64
ER source + moderate lab procedures (31–60)	Emergency admission	86.9%	1.64
ER source + med. change not changed	Emergency admission	86.5%	1.63
ER source + no readmission	Emergency admission	85.9%	1.62
ER source + many diagnoses (9+)	Emergency admission	85.7%	1.62
ER source + no procedures	Emergency admission	85.6%	1.61

Note: For the <30 minority class, lift is more informative than confidence due to the low base rate (11.17%).

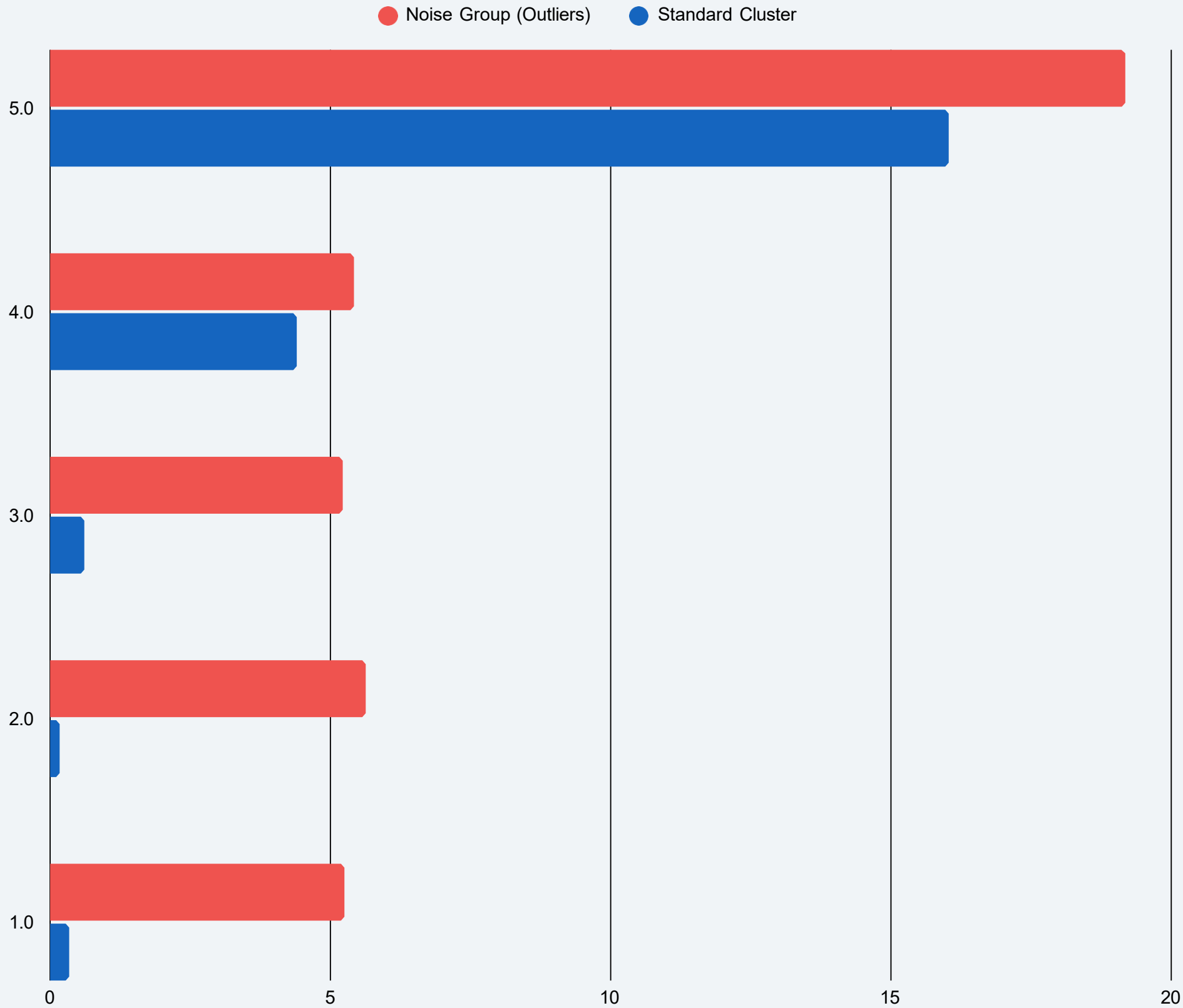
04 Clustering — DBSCAN

Parameters & Justification

min_samples	16	$2 \times d = 2 \times 8$ (Ester et al., 1996)
eps	2.0	Elbow of k-distance graph at ~100,000th point
Features	8 numerical	All numerical utilization features
Scaling	StandardScaler	Required for distance-based methods



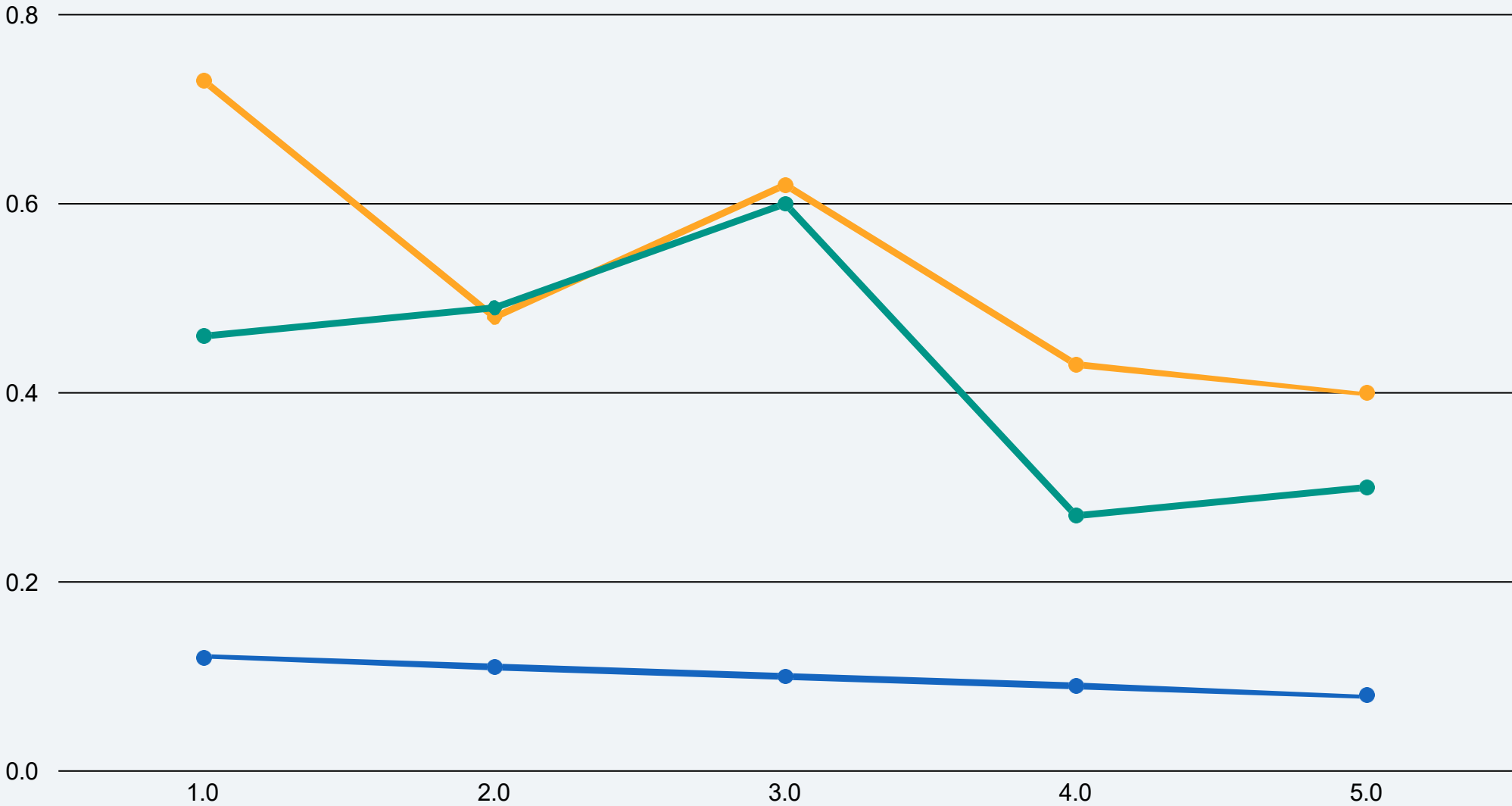
Outlier Patient Profile vs. Standard Cluster



04 Clustering — Hierarchical (Ward Linkage)

Linkage Method Comparison

Linkage	Silhouette	Calinski-H.	Min. Cluster	Viable?
Average	0.7275	27.37	0.04%	✗ Degenerate
Complete	0.4613	42.94	0.04%	✗ Degenerate
Ward ✓	0.1086	151.05	5.80%	✓ Balanced



Clinical Cluster Profiles (k=3, Ward)

Cluster 0 — 78.24% (1,956)

Routine Older Chronic-Care

Stay: 3.6 days | 14 medications
Circulatory diagnoses dominant
Lowest readmission risk

Cluster 2 — 15.96% (399)

Medication-Intensive Inpatients

Stay: 8.3 days | 25 medications
52% medication changes
14% high A1C rate

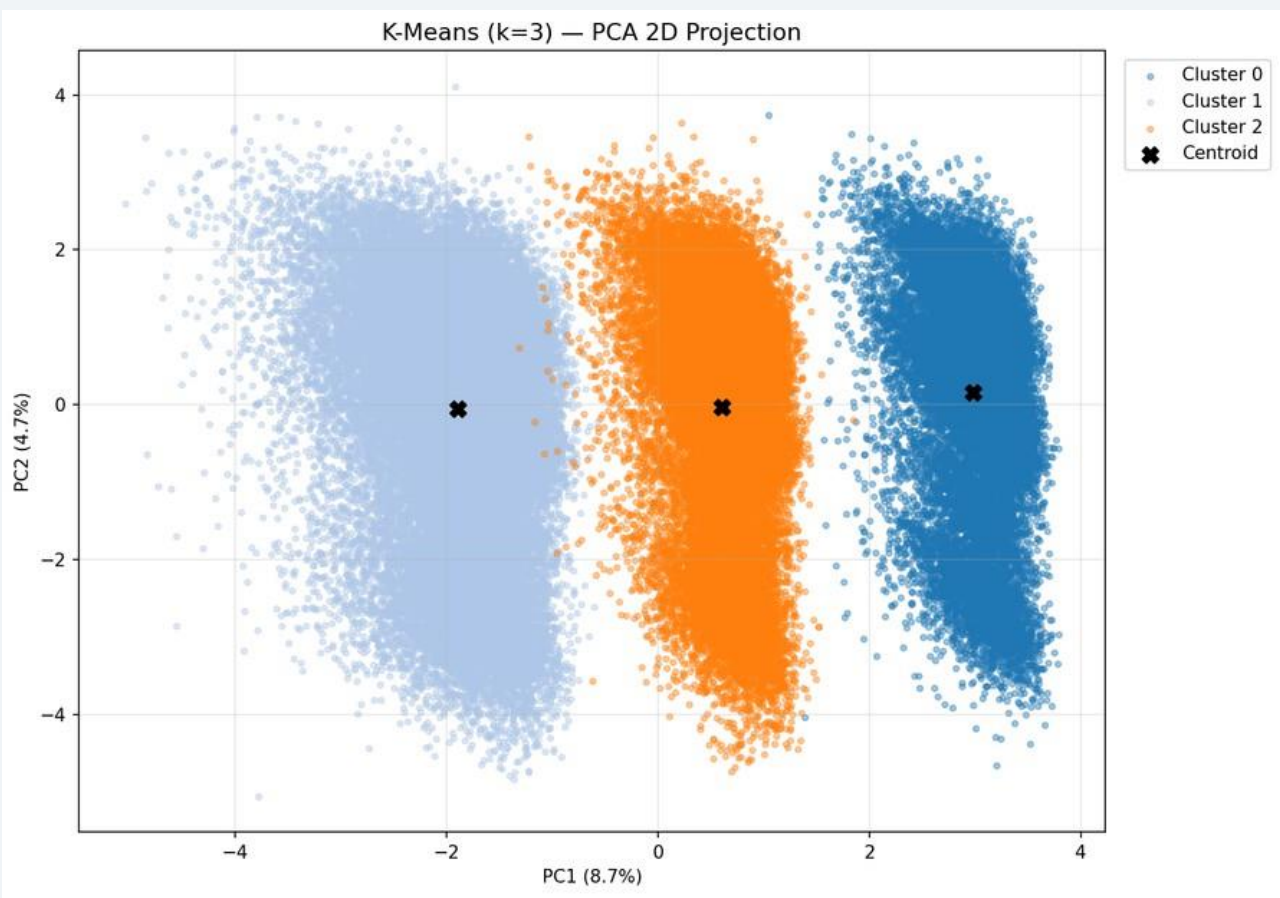
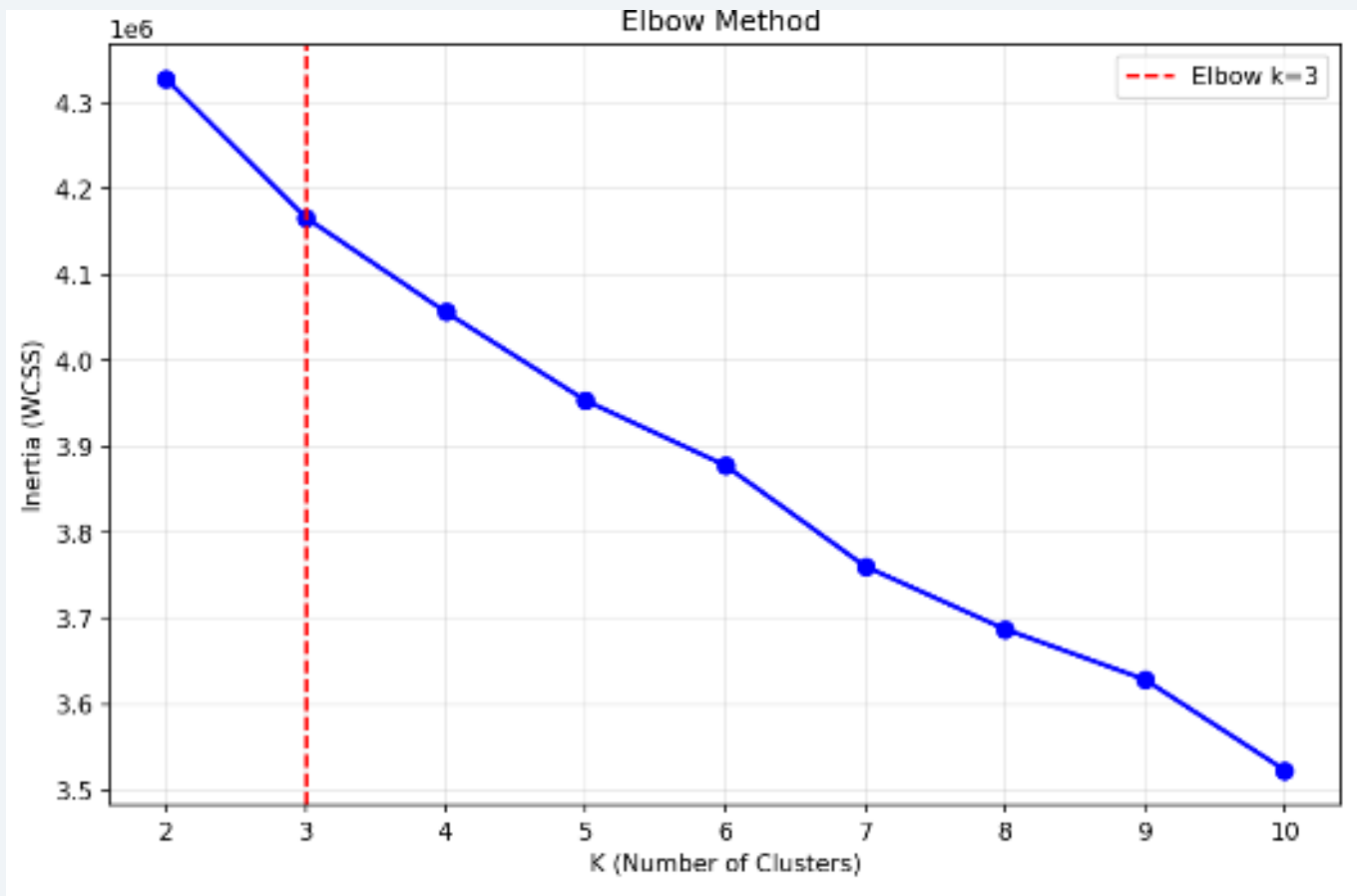
Cluster 1 — 5.80% (145)

Recurrently Admitted Acute-Care

1.70 prior ER visits | 4.25 inpatient hist.
Endocrine/Metabolic diagnoses
Highest readmission risk

04 Clustering — K-Means (k=3)

k=3 selected: Elbow method shows inflection at k=3. k=2 had best Silhouette (0.1144) and Calinski-Harabasz (7,861), but k=3 provides better clinical interpretability. Low silhouette values overall reflect naturally overlapping patient groups.



Cluster 0 Standard Care

Insulin ordinal: 0.00 |
Stay: 4.06d | Meds: 13.26
Well-controlled diabetics, no active insulin

Cluster 1 High-Intensity

Insulin ordinal: 1.31 |
Stay: 4.75d | Meds: 18.21 | Labs: 44.46
Active pharmacological intervention needed

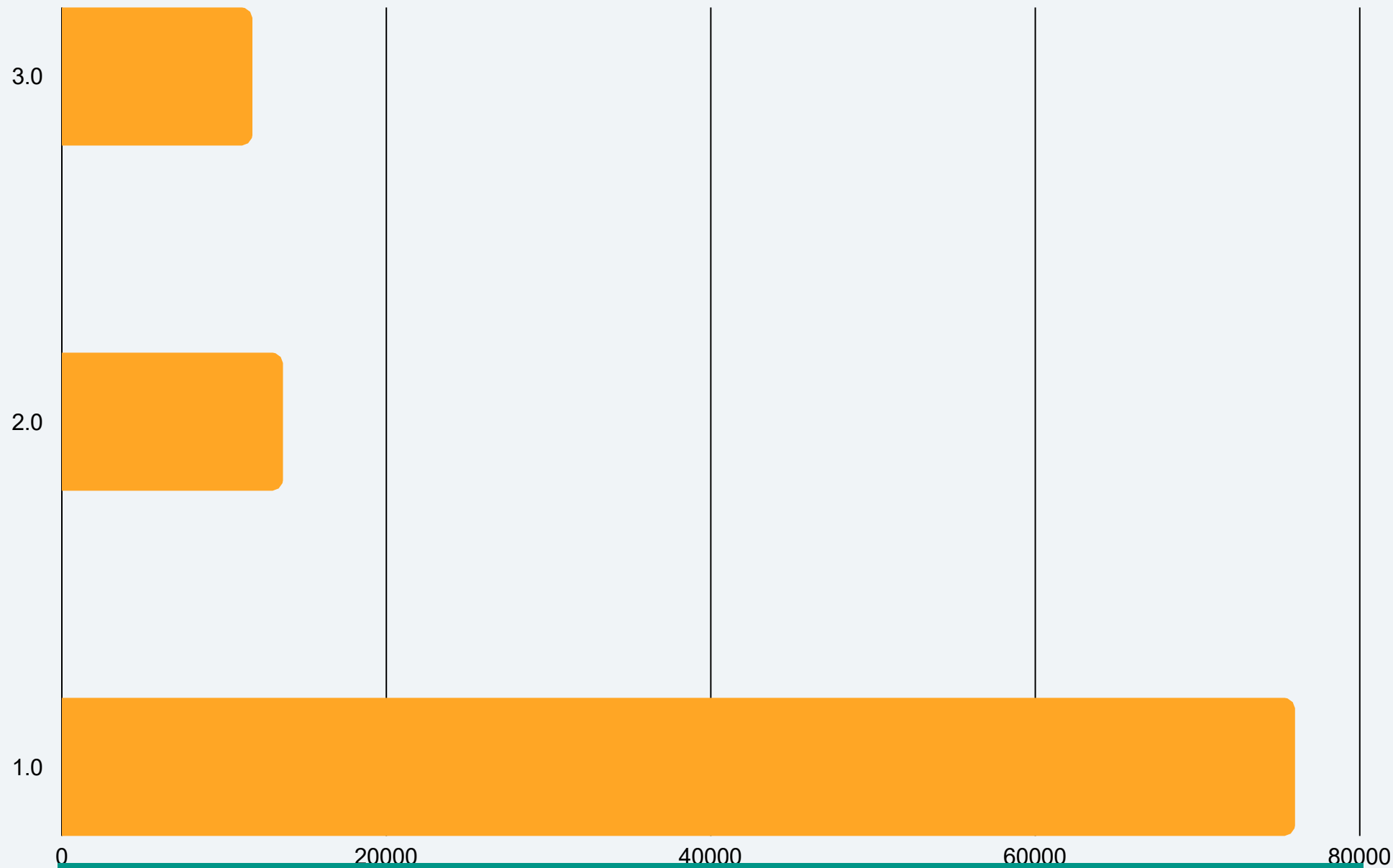
Cluster 2 Moderate-Insulin

Insulin ordinal: 0.52 |
Stay: 4.13d | Meds: 14.89
Transitional — between routine & high-intensity

04 Clustering — Gaussian Mixture Models (GMM)

Why GMM over K-Means?

- Hard vs. Soft Boundaries**
K-Means applies rigid cluster boundaries; GMM uses probabilistic (soft) clustering based on hidden densities.
- Medical Data Overlap**
Patient profiles naturally overlap — GMM captures these grey areas in clinical reality that rigid methods miss.
- Probability Scores**
GMM computes the probability of each patient belonging to each cluster, capturing uncertainty.
- Feature Engineering**
GMM cluster probabilities will be used as additional soft features in downstream predictive models.

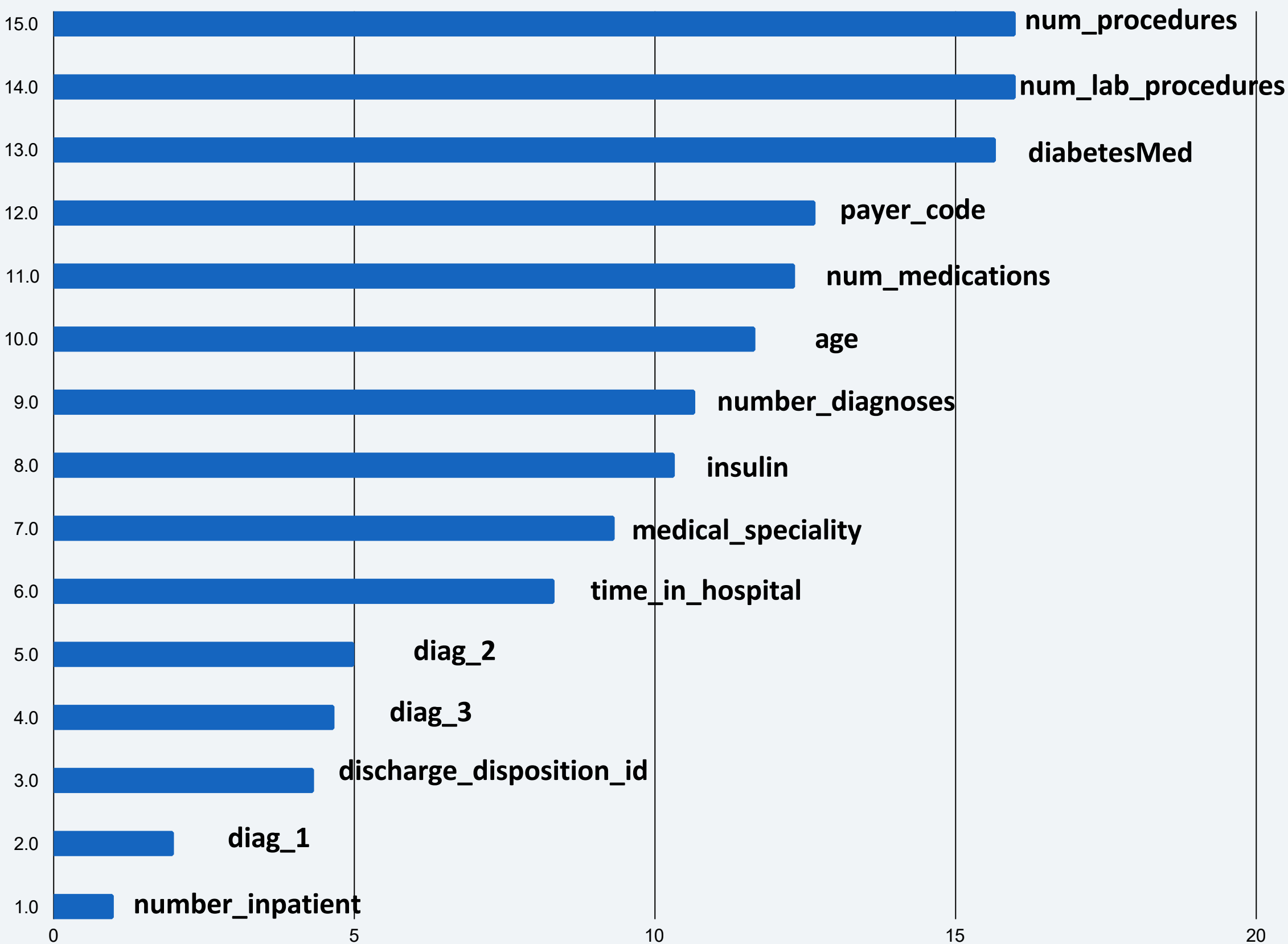


Key Insight

The significant cluster overlap in GMM's PCA projection accurately reflects clinical reality: diabetic patient conditions and resource utilizations form a continuous, overlapping spectrum. This nuanced probabilistic understanding makes GMM outputs valuable as soft features for downstream predictive models.

05 Classification — Feature Selection

3 methods combined: Mutual Information (MI) + Chi-Square + Random Forest → Average Rank. Lower rank = stronger feature.



Mutual Information

Highlights number_inpatient, diagnosis codes, and discharge_disposition_id as top informers — captures both linear and non-linear dependencies.

Chi-Square

Elevates insulin and diabetesMed into high positions, indicating medication regimens are associated with class-level differences in readmission.

Random Forest

Uniquely promotes num_lab_procedures to rank 3, suggesting it contributes useful decision-tree splits beyond standalone statistical dependency.

Top Feature: number_inpatient (Rank 1 in ALL 3 methods)

05 Classification — k-NN (Early Readmission)

Experiment	Features	Distance	Best k
Exp 1: Top 10 + Euclidean	10	Euclidean	3
Exp 2: Top 20 + Metric Comparison	20	Euclidean	5

Test Set Results

Experiment	Accuracy	Precision	Recall	F1	ROC-AUC
Exp 1: Top 10 + Euclidean	0.8628	0.1851	0.0671	0.0985	0.5393
Exp 2: Top 20 + Metrics	0.8783	0.2225	0.0358	0.0616	0.5450

Confusion Matrix Comparison

	TN	FP	FN	TP
Exp 1	17,343	669	2,113	152
Exp 2	17,729	283	2,184	81

Verdict

k-NN struggles with class imbalance — the algorithm naturally skews toward the majority class. Both experiments show ROC-AUC barely above 0.5, indicating weak discrimination.

Key Insight: For imbalanced datasets, accuracy alone is misleading. Recall is the critical metric when missing a high-risk patient has clinical consequences.

05 Classification — Random Forest

Experiment 1 — Baseline	
Feature set:	All 46 features
n_estimators:	100
max_depth:	None (fully grown)
min_samples_leaf:	10
class_weight:	balanced

Experiment 2 — Tuned ★	
Feature set:	Top 20 by RF importance
n_estimators:	300
max_depth:	15
min_samples_leaf:	10
class_weight:	balanced

Metric	Exp 1 — All Features	Exp 2 — Top 20 Features
Feature count	46	20
Accuracy	0.8464	0.7763
Precision	0.2563	0.2098
Recall	0.1974	0.3625
F1-Score	0.2230	0.2658
ROC-AUC	0.6599	0.6615
PR-AUC	0.2115	0.2093

Key Result

Tuned RF (Exp 2) achieves 36.3% recall vs 19.7% baseline — nearly 2× improvement.

ROC-AUC stays stable at 0.66, confirming overall discriminative power is preserved.

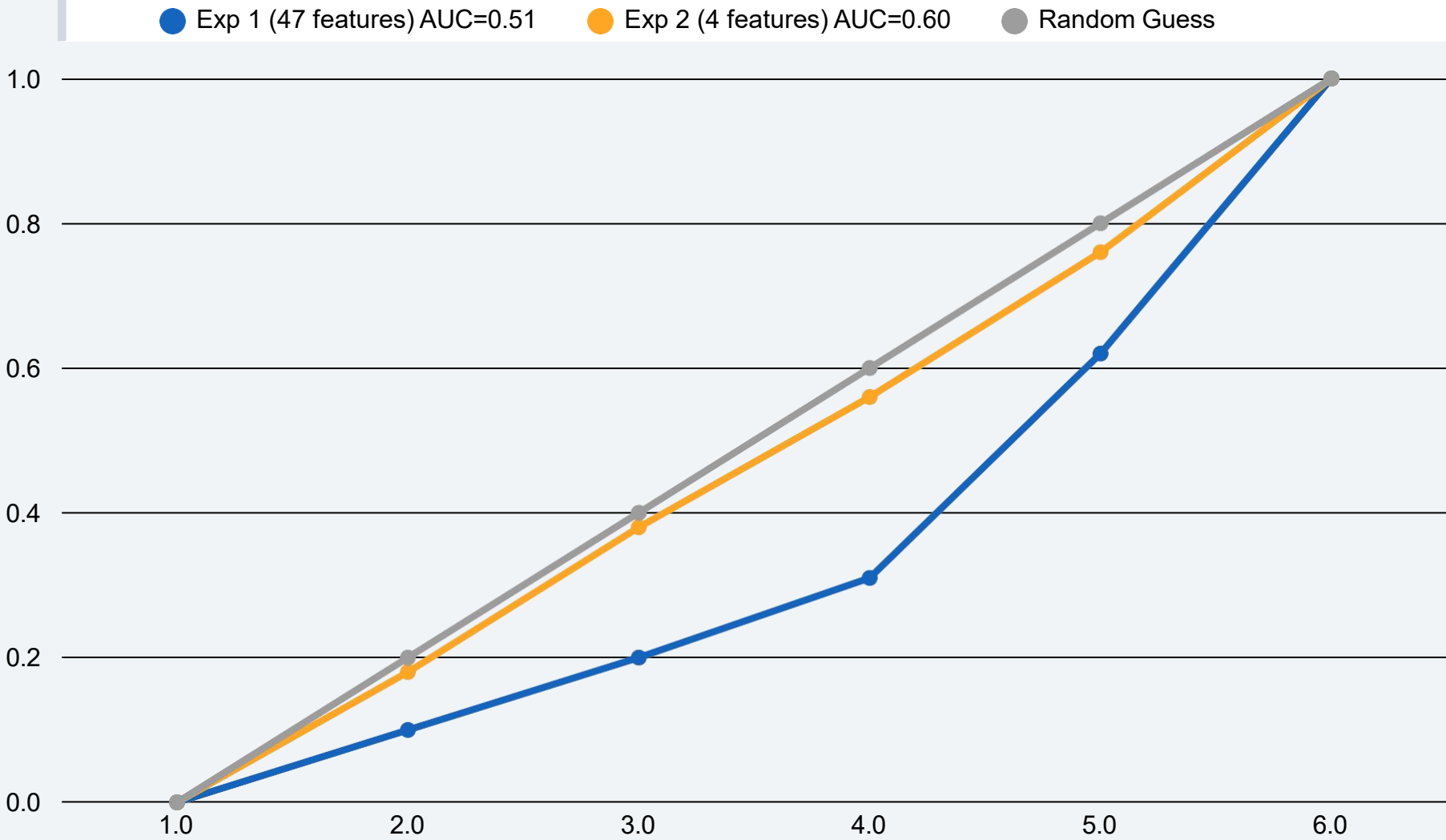
Feature selection (46→20) + depth constraint (max_depth=15) leads to a more clinically useful model that misses fewer high-risk patients.

Top predictors: number_inpatient (0.122), diag_1 (0.090), num_lab_procedures (0.086).

05 Classification — Naive Bayes

Feature Engineering (Exp 2)

- total_visits: outpatient + emergency + inpatient
- total_interventions: medications + lab + procedures
- + 2 additional high-signal features from ANOVA
- 4 features total — stripped of all noise



Metric	Exp 1 (47 features)	Exp 2 (4 features)	Change
Accuracy	82.90%	87.51%	+4.61%
ROC-AUC	0.51	0.60	+0.09
Precision	7.19%	27.33%	+279%
Recall	4.46%	7.11%	+59%
F1-Score	0.0551	0.1128	+104%

Noise in Medical Data

47 features overwhelmed the model (AUC 0.51 = random). Just 4 engineered features raised AUC to 0.60 — less data, higher quality = better predictions.

False Positives Slashed

Baseline: 1,303 FPs (wasted hospital resources). Engineered model: 428 FPs (–67%). Precision jumped from 7% to 27%.

Recall Limitation

Recall remains low (7.1%). 2,104 FNs remain. SMOTE or threshold tuning needed to catch more high-risk patients.

05 Classification — Decision Tree

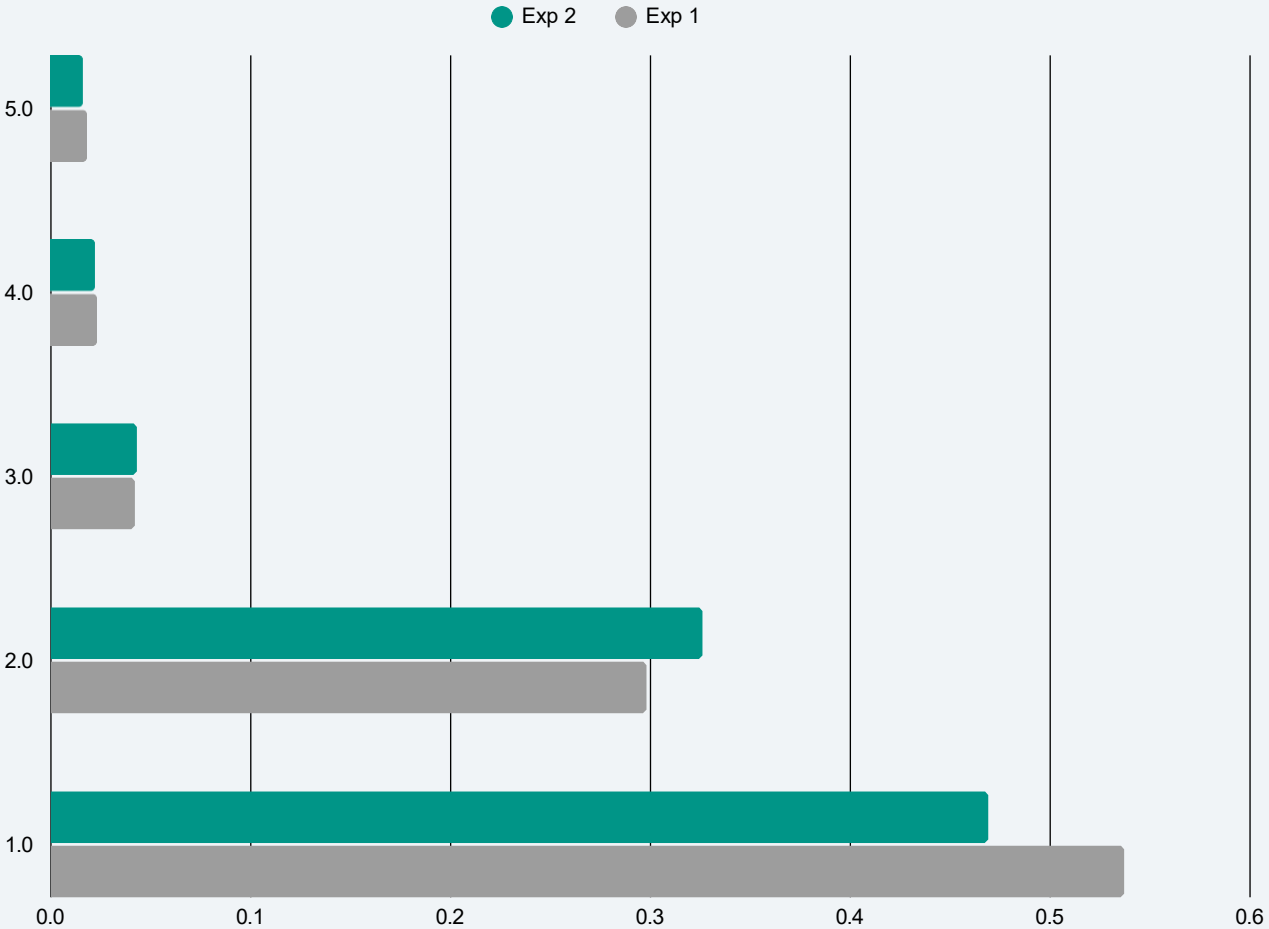
Exp 1 — All Original Features

- 46 features
- Criterion: Gini
- Max Depth: 6
- Min Leaf: 100

Metric	Exp 1 — All Orig.	Exp 2 — Engineered
Accuracy	0.6164	0.5959
Precision	0.1682	0.1651
Recall	0.6172	0.6450 ✓
F1-Score	0.2644 ✓	0.2629
ROC-AUC	0.6537	0.6608 ✓
PR-AUC	0.2037	0.2109 ✓

Exp 2 — Engineered Clinical Subset ✓

- 28 features (20 + 8 engineered)
- Criterion: Gini
- Max Depth: 8
- Min Leaf: 200

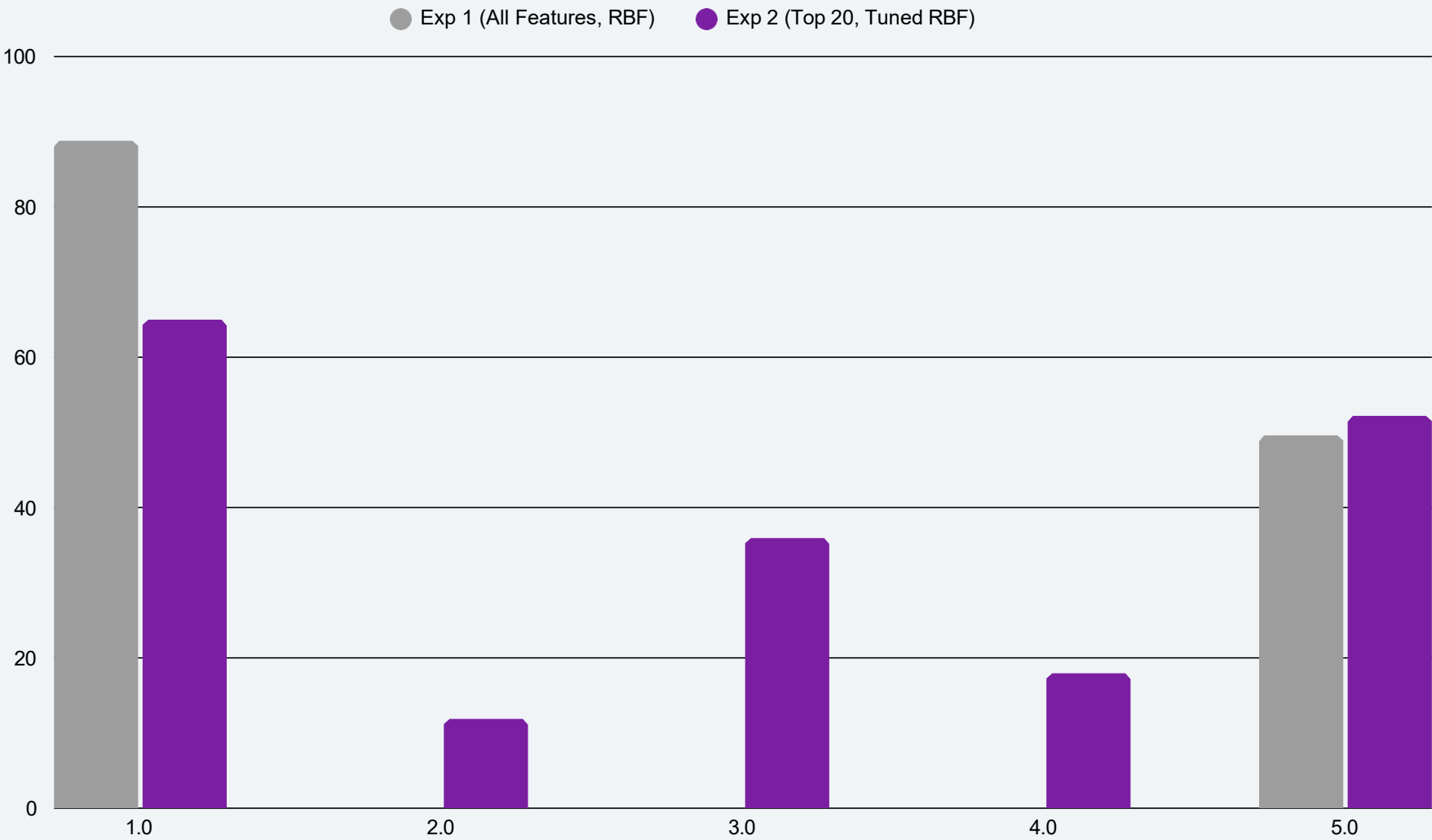


Exp 2 achieves higher recall (64.5%), ROC-AUC (0.661), and PR-AUC (0.211), confirming that engineered features help detect more actual early readmission cases. `number_inpatient` and `discharge_disposition_id` dominate both experiments, validating prior feature selection findings.

05 Classification — Support Vector Machine (SVM)

SVMs find the optimal hyperplane maximizing the margin between classes. For non-linearly separable data, the RBF kernel maps data to higher dimensions. Training was subsampled to 10,000 records due to $O(n^2)$ complexity.

Experiment	Feature Subset	Kernel	C (Penalty)	Gamma
Experiment 1	All Features (46)	RBF	1.0 (Default)	'scale' (Default)
Experiment 2	Top 20 Features	RBF	0.5 (Tuned)	0.01 (Tuned)



	TN	FP	FN	TP
Exp 1	18,012	0	2,265	0 ✗
Exp 2	11,976	6,036	1,451	814 ✓

Exp 1 Complete Failure

Default SVM predicted zero positive cases (TP=0, Recall=0.0). Unable to establish an effective separation hyperplane for the minority class in high-dimensional space.

Exp 2 Improvement

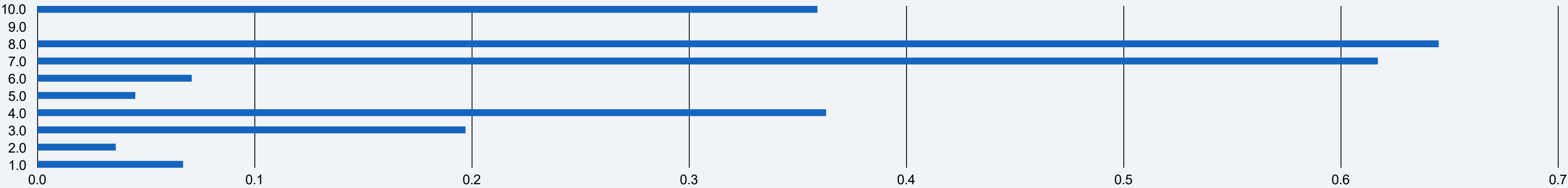
Feature selection & tuning raised recall to 35.9%, correctly identifying 814 early readmissions. However, 6,036 false positives make this unreliable clinically.

Conclusion

SVM ROC-AUC (0.522) is barely above random. F1-score (0.179) is lower than Random Forest (0.266) and Decision Tree (0.264). SVM is less suited for this imbalanced task.

05 All Models — Performance Comparison

Model	Experiment	Accuracy	Precision	Recall	F1	ROC-AUC
k-NN	Exp 1 (Top 10 Euclidean)	0.8628	0.185	0.067	0.099	0.539
k-NN	Exp 2 (Top 20 Metrics)	0.8783	0.223	0.036	0.062	0.545
Random Forest	Exp 1 (All features)	0.8464	0.256	0.197	0.223	0.660
Random Forest	Exp 2 (Top 20 Tuned)	0.7763	0.210	0.363	0.266	0.662
Naive Bayes	Exp 1 (47 features)	0.8290	0.072	0.045	0.055	0.510
Naive Bayes	Exp 2 (4 features)	0.8751	0.273	0.071	0.113	0.600
Decision Tree	Exp 1 (All features)	0.6164	0.168	0.617	0.264	0.654
Decision Tree	Exp 2 (Eng. subset)	0.5959	0.165	0.645	0.263	0.661
SVM	Exp 1 (All features)	0.888	N/A	0.000	0.000	0.496
SVM	Exp 2 (Top 20 Tuned)	0.650	0.119	0.359	0.180	0.522



05 Best Model — Decision Tree (Exp 2 — Engineered Subset)

0.645

Recall

0.263

F1-Score

0.661

ROC-AUC

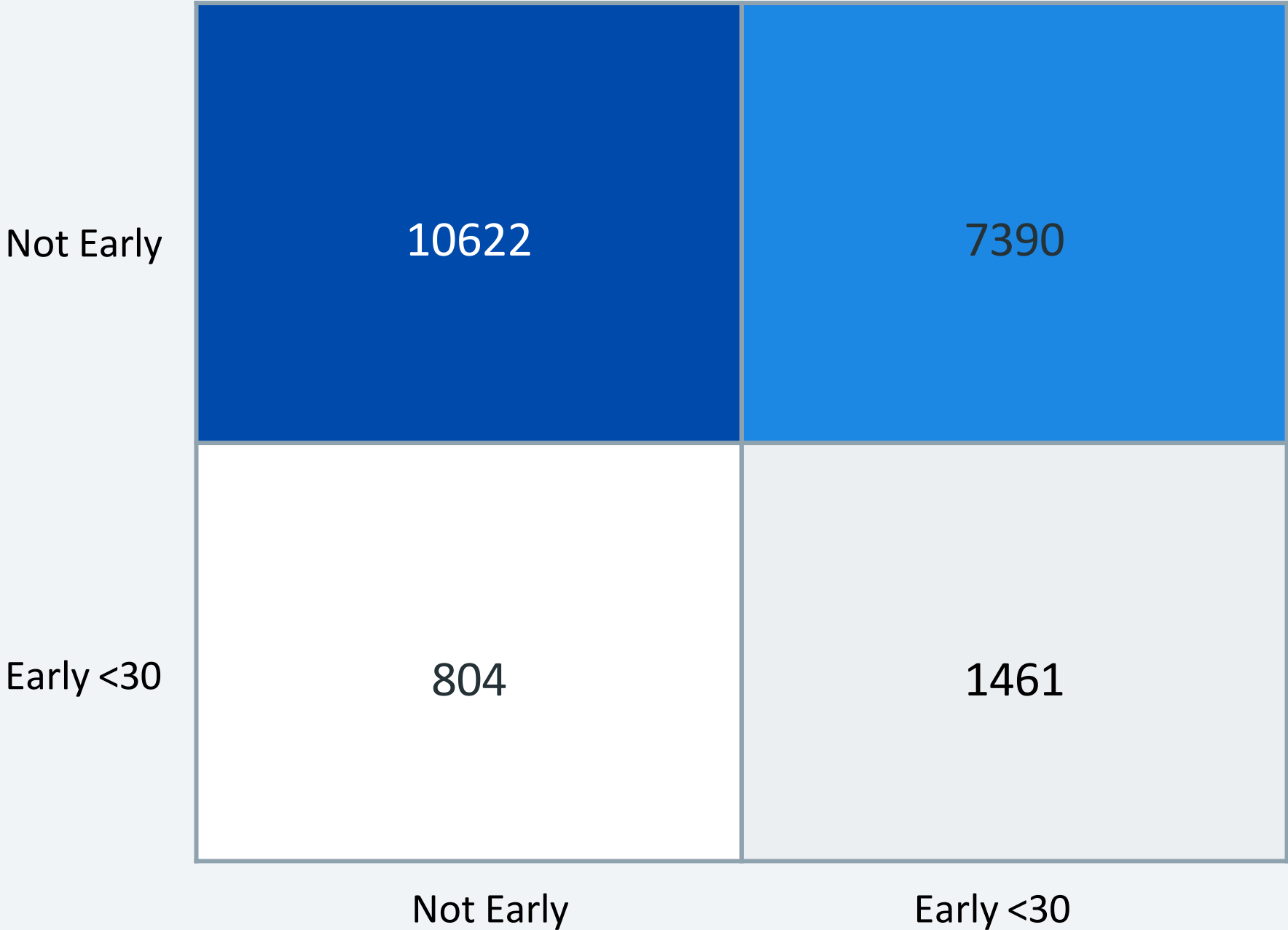
0.211

PR-AUC

	Predicted: 0	Predicted: 1
Actual: 0	TN: 10,622	FP: 7,390
Actual: 1	FN: 804	TP: 1,461

Statistical Significance Note: Decision Tree Exp 2 achieves the highest recall (0.645) and PR-AUC (0.211) across all models, and ties for top ROC-AUC (0.661). In a clinical setting where missing a high-risk patient carries real consequences, its recall advantage over Random Forest Exp 2 (0.363) and all k-NN/SVM variants is decisive. Precision remains low (0.165) due to inherent class imbalance (11.17% minority) — a known trade-off when optimising for recall on this dataset.

Confusion Matrix - Experiment 2



06 Conclusions

Descriptive & Clustering Insights

- Older patients (60-90) dominate; emergency admission is the primary pathway.
- number_inpatient is the single strongest predictor — even before modeling.
- Apriori: prior inpatient history (2+ visits) lifts <30 readmission risk $\sim 1.91\times$.
- DBSCAN: 0.6% truly atypical high-utilization patients form a clinically distinct outlier group.
- Hierarchical / K-Means / GMM all identify 3 patient tiers: routine, moderate, high-complexity.

Predictive Analysis Insights

- Decision Tree achieves highest raw recall (0.64) .
- Feature engineering critical for Naive Bayes: 47 $\rightarrow$ 4 features improved ROC-AUC from 0.51 $\rightarrow$ 0.60.
- SVM completely failed in default config (TP=0); tuning helped recall (0.36) but high FP rate.
- Class imbalance (11.17% minority) is the central challenge across all models.

Practical Clinical Insights

- Patients with 2+ prior inpatient visits need targeted discharge planning and follow-up.
- Skilled nursing facility discharge $\rightarrow 1.31\times$ higher readmission risk.
- High medication burden + multiple diagnoses $\rightarrow$ intensive monitoring required.
- Automated risk scoring using RF model could flag patients for intervention at discharge.

06 Limitations & Future Improvements

Limitations

Class Imbalance

All models struggle with the 11.17% minority. SMOTE, ADASYN, or cost-sensitive learning could significantly improve recall across all classifiers.

Race Imputation

1,764+ missing race values filled by class distribution — may introduce subtle demographic bias into downstream features.

Hierarchical Scale

Hierarchical clustering fitted on 2,500-record sample only, not the full 101K dataset, which may miss population-level structure.

kNN & SVM Scale

k-NN used 5,000 training records and SVM 10,000 due to $O(n^2)$ complexity — both may miss patterns visible in the full data.

No Temporal Analysis

Dataset lacks temporal sequencing; readmission prediction could benefit significantly from time-series and sequential diagnosis features.

Future Improvements

Apply SMOTE / ADASYN

Address class imbalance for better minority-class recall across all models.

Neural Networks

Deep learning with attention mechanisms for readmission prediction.

XGBoost / LightGBM

Gradient boosting methods are likely to outperform all traditional classifiers tested.

Temporal Features

Engineer time-since-last-visit and sequential diagnosis features.

Ensemble Stacking

Combine RF, DT, and GMM probabilistic features into a meta-learner.

Threshold Tuning

Optimize decision thresholds per clinical cost-benefit analysis.